
SKILLS

- **ML Systems:** model serving, inference APIs, model validation, monitoring, retraining workflows, reproducible research
- **Deep Learning:** PyTorch, TensorFlow, HuggingFace, transformers, embeddings, inference pipelines, batch/online scoring
- **MLOps:** feature pipelines, model stores, eval harnesses, drift detection, CI/CD, experiment tracking, model risk controls
- **Optimization:** profiling, latency SLOs, quantization concepts, runtime tuning, caching, parallelism, hardware-aware deployment
- **Data Platforms:** Kafka, Airflow, Snowflake, PostgreSQL, Redis, Elasticsearch, ETL, schema design, feature stores
- **Engineering:** Python, Go, TypeScript, SQL, REST APIs, microservices, Docker, Kubernetes, Terraform, Linux
- **Reliability:** OpenTelemetry, ELK, CloudWatch, distributed tracing, alerting, production debugging, incident response

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect – AI / ML Systems Nov 2022 – Present
 - **ASB Assist – GenAI & Retrieval System:** Building an LLM + RAG system on RBC's enterprise blueprint platform with retrieval pipelines, context assembly, prompt/version comparison, citation validation, audit trails, guardrails, and offline evaluation before production rollout.
 - **MLOps & Evaluation Infrastructure:** Designed model-evaluation workflows for regulated AI use cases: regression test sets, answer-quality metrics, traceability, Responsible AI controls, approval checkpoints, and reproducible iteration from prototype to production.
 - **AI/ML Observability Platform:** Architected telemetry platform using OpenTelemetry, Vector, Fluent Bit, Kafka, and Logstash ingesting 150 TB/day across 10,000+ systems; standardised schemas enabling model monitoring, drift signals, latency SLOs, anomaly detection, and AIOps workflows.
 - **0-to-1 Production Platform Delivery:** Led production delivery of a patent-pending Node.js/TypeScript/React platform with rules-based recommendation engine, real-time data rendering, secure service matching, and auto-certification; wrote 99.6% of production code and 100% of core logic.
 - **Technical Leadership & Research-to-Production:** Led 5 engineers through technical planning, code reviews, ADRs, testing expectations, runbooks, operational handoff, and stakeholder alignment with product, risk, security, architecture, and infrastructure teams.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - **Production ML Inference Pipeline:** Integrated transformer-based document classification for medical and insurance workflows; deployed inference APIs on AWS EKS with monitoring, HIPAA-aligned audit logging, and production feedback loops, reducing manual review by 60%.
 - **Feature & Trust-Score Workflows:** Designed validation and trust-score pipelines on Hyperledger Fabric for real-time premium verification and fraud-resistant claims processing; joined as first engineer and scaled platform to 20+ engineers before acquisition.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
Software Developer Co-op Apr 2022 – Nov 2022
 - **Production Software Systems:** Migrated federal tax platform UI from Apache Struts to React/TypeScript and contributed to backend APIs serving millions of Canadians with accessibility, quality gates, maintainability, and secure deployment standards.
- **Dalhousie University** Halifax, Canada
Research & Teaching Assistant Jan 2021 – Aug 2022
 - **Algorithms, Systems & Cloud:** Researched call-stack decision algorithms and Python tracing tools for compiler/runtime analysis; TA for Algorithms, Software Development, and graduate Advanced Cloud Computing.

PROJECTS

- **Enterprise Document Q&A with LLM Evaluation Harness:** LangChain + pgvector RAG over messy document corpora with offline eval sets, regression metrics, prompt/version comparison, citation validation, tracing, and quality dashboards.
- **Flight Price Optimization Agent:** DQN/DDQN reinforcement-learning agent in Python for dynamic pricing; modelled demand curves on historical flight data and improved simulated outcome by 18% over baseline.
- **Portfolio Risk Optimization Framework:** Python research framework for market data ingestion, backtesting, Monte Carlo simulation, risk-adjusted signal evaluation, and reproducible quantitative experiments.

EDUCATION

- **Dalhousie University** Halifax, Canada
MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing Jan 2021 – Aug 2022
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) Completed 2024
- **Gujarat Technological University** Ahmedabad, India
B.E. Computer Engineering, GPA: 3.92/4.0 Aug 2016 – Aug 2020